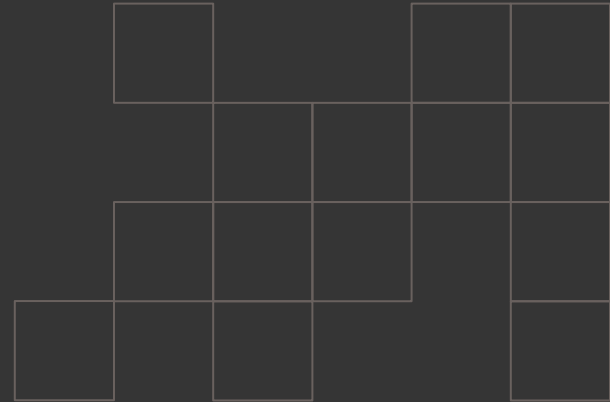


ASME 2026-2027 Overview



What is ASME?



- **American Society of Mechanical Engineers**
- **The Student Design Competition is a National Level Design and Manufacturing Challenge**
- **ASME** is a great opportunity to get hands on experience, Big Engineering Companies such as **Lockheed Martin** actively look for **ASME** in your Resume

Time Commitment Project Documentation Team collaboration.

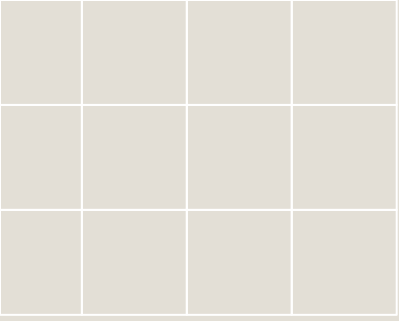


- ASME Is a large time commitment.
- Good time management and communication is key to be successful
- School and personal matters come first, if you are unable to work on the project after being selected you will be put on reserve .
- Time spent working on the project and Efficiency will be taken into account for the travel team
- The Project must be fully documented ; every concept sketch, design in CAD, and physical prototype iteration.
- Each team must collaborate and communicate with each other on a consistent basis.



Summarized Rule Set

- **Robot limits:** The Robot must fit inside a 40x40x40 cm cube and can't weigh more than 15kg. At the start of a Run the Robot is allowed to expand its size to 75 cm horizontally and 120 cm in height
- **Power:** Onboard electricity only. Batteries and controls are 24V nominal or less. Pneumatics are limited to 60 psi and must have a pressure gauge and release valve, and if hydraulics are used we need to use 100 psi or less and use non-flammable, food-grade hydraulic fluid. Stored mechanical energy is allowed as long as the Robot is able to restore the mechanical energy during operation.
- **Conduct:** Do not touch the Robot during a run or leave the field, cause damage, interfere, play defense, or grapple another robot
- **The Robot must pass the inspection checklist:**
 - Only one robot per team.
 - fits within the team's 40 x 40 x 40 cm cube. Expansion within the limits.
 - Weighs 15kg or less.
 - Electric power only; no combustion. Battery 24V nominal or less; undamaged; main switch (to turn off the Robot immediately).
 - Fire-safe charging bag present.
 - Stored-energy pre-load can be restored by the Robot itself.
 - No flight. One control link on a license-free band; can be disabled.
 - Pneumatics: visible gauge reads 60 psi or less when charged; relief valve present.
 - Ascent anchor point present, accessible, and rated for the Robot's full weight. Team identifier shown; flat area for Pair marker. No prohibited or hazardous features; safe edges.
 - Team has its own eye protection marked Z87 or EN166 or equivalent marking on the frame or lens. **End of inspection list.**



**For this years ASME
Project we need to create
a robot to perform 5
different tasks.**



The Sprint



- Cross a course of Gates and Hurdles as fast as possible. The Robot may either go over a Hurdle or pass it on the open side.
- **How to perform this task:** When you start the Run the four gates must be taken in order, on alternating sides, and both Hurdles should be crossed by driving over them. **IF we can not go** over a Hurdle we get a **penalty.**

The Lift



- **Place calibrated Dumbbells onto a three-tier podium.** The **heavier Dumbbells** placed on a taller podium **score more points.**
- **How to perform this task:** The robot must carry Dumbbells from the loading zone and place them on the podium. Dumbbells may be stacked to score more points on a higher podium. When the final buzzer goes off the Dumbbells must be stable and resting on the podium without the help of the robot. If a dumbbell falls off the stack after the buzzer rings the whole stack is voided and we receive no points. The heaviest Dumbbell will be 4kg

The Cycle



- The objective is to **collect balls** and **score them through goals**.
- **How to perform this task:** The Robot has to collect balls and score them through the goals. A ball counts when it passes fully through a goal opening. There are two goals the High goal and the Low goal. The high goal is seated 1 meter above the play field. The High goal gives more points than the Low goal.

The Ascent



- **Climb a wall.** The **higher** the Robot is **holding** at the **buzzer**, the **more it scores**, and **climbing on its own** is worth more points. The summit is 2 meters in the air and the starting zone is on the floor in-front of the wall.
- **How to perform this task:** The Robot has to climb to 3 different rungs at different heights, the lowest rung is .5 m from the ground, and the highest is 1.5 m the maximum amount of points is scored at the summit which is 2+ m from the ground.

The Marksman



- **Land bean bags** on a ringed target as **accurately** as possible, **with a loader** feeding one bean bag at a time.
- **How to perform this task:** The Robot need to basically play cornhole and launch bean bags onto the target as accurately as possible. Launchers are not allowed to be pretensioned before the Run.

- There are **three teams** in ASME. **Mechanical, Electrical, and Programming.**
- **Team members** will be **decided** through a **vetting process**. In the vetting process you will have to **design a functioning mechanical part, circuit, or program** to be eligible **to join** one of the three teams



How the Vetting Process will work for the Mechanical team:

- **There will be a poll on discord with the five disciplines, you will choose the two that interest you the most. Once the polls end you will be assigned one of the disciplines you have chosen, you will then need to design a functioning part from the discipline assigned to you.**
- **You will be given 1 month in total to complete the design and have it fabricated. You should have your idea sketched out as quickly as possible preferably within a week, you'll have to send your sketched out idea to Gavin on discord or to his student email (gxk46800@email.vccs.edu) . After you finish sketching your idea, you will begin fabricating your part (through 3D printing)**
- **The Deadline is 10/02/26 during the clubs meeting**



You will be graded on:

- 1. Competition Value / Quality of Idea [25]**
- 2. Functionality [20]**
- 3. Engineering Reasoning [20]**
- 4. Testing & Improvement [15]**
- 5. Mechanical Design Quality [10]**
- 6. Project Effort, Time Management & Understanding[10]**

